

AER 1515: Perception For Robotics

Project Instructions

Fall 2025

September 1, 2025

The project component of the course will be on a topic of your choosing, agreed to with Prof. Waslander upon receipt of the max. two page project proposal. Groups of up to two students are allowed, and the project content must be related to some aspect of the course, and as such must have a robotic perception component to it. Initial results will be required mid-project, then a video describing the project and demonstrating its operation will be required toward the end of the semester and a final report in two column IEEE conference paper format not to exceed 6 pages (+ references) will complete the project.

Marking:	Proposal	5
	Initial Results	5
	Video	15
	Report	35
	Total:	60% of course mark

1 Proposal Instructions

The proposal should be 2 pages maximum and should define the problem you plan to work on and its relation to the course content. There is no doubt you will have to read online and textbook materials to determine the direction of the project. The following topics should be covered in the proposal:

- Outline problem, applications
- Relationship to course
- Solution method
- Datasets to be used, ground truth availability
- Goals/Timeline for semester

There is no mark for the proposal, but it is the main chance to get feedback on the direction of the project, so be as specific and technical as possible in two pages. Upon receipt of your proposal feedback, I will make time available to set meetings to discuss progress and direction in more detail.

2 Initial Results

The initial results submission will extend the proposal document to a maximum of 4 pages, submitted in PDF form, as follows:

- The same five sections will be included with updated descriptions of the problem, relationship to course, solution method, datasets used and an updated timeline, based on feedback from Prof. Waslander and any course corrections that have occurred.
- An additional section after datasets that presents initial results in the form of evaluation metrics of a method on the selected data and/or visualizations of the method performing on input data (images/lidar, with ground truth labels if applicable).

The marking will be divided into two, 20% for proposal refinement writing/presentation quality, and 80% for progress-to-date in generating initial results that demonstrate your team has implemented and begun experimentation with your chosen perception method.

3 Video Instructions

The video is intended to be a showcase of your work for the whole class, and will be due for the last week of lecture. Results must be included, demonstrating the successful implementation of your project on realistic simulated or real data, and the video should present the problem, method used for solution and results. It is not necessary to record a voice-over if the video visuals are sufficiently detailed to explain the core ideas effectively, but it is often easier to convey detailed content with a scripted recording. The maximum duration for the video is 5 minutes.

The video will be evaluated as follows:

1. Project Motivation/Applications (20%): Is there a believable application for the work, does it relate to the course content, is there a need that is fulfilled by the project?
2. Quality of Technical Content (30%): Are the concepts employed clearly understood, explained and applied correctly, is the work an extension of the theory presented in class, are derivations correct, reasoning valid, content of use and well thought out?
3. Overall Completion of Objectives (30%): Did the team set reasonable goals and achieve those goals this semester, was sufficient effort expended to achieve those goals?
4. Presentation Quality (20%): Was the video clear, concise, did it document progress well?

4 Report Instructions

The final report should be six pages of content plus references in two column IEEE format. Summarize the results of your project, as if in preparation for a robotics conference. You can follow the instructions for an ICRA conference submission, as provided here:

Latex: <http://ras.papercept.net/conferences/support/tex.php>

Word: <http://ras.papercept.net/conferences/support/word.php>

The following main topics should be covered, so that the report is in the form of a complete, conference ready paper.

- Explain motivation, related work in introduction
- Define problem clearly
- Present algorithms, theory clearly and concisely.
- Present simulation results or experimental results that clearly demonstrate the benefits of your methods (comparisons are preferred).

- Summarize and suggest future work

Be sure to focus on interesting extensions and additions that your project entails. There is no need to re-derive what is in the course notes in the report. The report will be evaluated as follows:

Category	Percent Weight
Introduction	(20%)
Motivation / Applications	
Related Work	
Quality of Technical Content	(50%)
Problem Formulation	
Algorithm Definition	
Extensions to Course Knowledge	
Completion of Objectives	(20%)
Functional Simulations	
Clear Presentation of Results	
Report Quality	(10%)
Formatting, spelling, grammar	
Images, labeling of axes	